

MATH3018 Assignment 7, due on Apr 16

Questions (1)-(3) use the same dataset. The correlation matrix of a $p = 6$ population is

$$\begin{pmatrix} 1 & & & & & \\ 0.505 & 1 & & & & \\ 0.569 & 0.422 & 1 & & & \\ 0.602 & 0.467 & 0.926 & 1 & & \\ 0.621 & 0.482 & 0.877 & 0.874 & 1 & \\ 0.603 & 0.450 & 0.878 & 0.894 & 0.937 & 1 \end{pmatrix}.$$

The following estimated factor loadings are obtained by the maximum likelihood method:

Standardized Variables	Estimated Factor Loadings $\hat{l}_{jk}$			
	Unrotated		Varimax Rotated	
	F_1	F_2	F_1^*	F_2^*
Z_1	0.602	0.200	0.484	0.411
Z_2	0.467	0.154	0.375	0.319
Z_3	0.926	0.143	0.603	0.717
Z_4	1.000	0.000	0.519	0.855
Z_5	0.874	0.476	0.861	0.499
Z_6	0.894	0.327	0.744	0.594

- (1) This question uses the unrotated estimated factor loadings. Find the maximum likelihood estimates of the followings:
 - (a) The communalities.
 - (b) The specific variances.
 - (c) The proportion of variance explained by each factor.
- (2) Is the proportion of variance explained by the factor F_1^* the same as that of F_1 ? How does the total proportion of variance explained by F_1^*, F_2^* compared to that of F_1, F_2 ? Provide some justifications for your conclusions.
- (3) Using the principal component method we also get a model with two factors F_1^{**} and F_2^{**} . Is the proportion of variance explained by the factor F_1^{**} the same as that of F_1 ? Which one is larger? How does the total proportion of variance explained by F_1^{**}, F_2^{**} compared to that of F_1, F_2 ? Provide some justifications for your conclusions.

(4) Establish the inequality

$$\left\| \mathbf{S} - \left(\hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\mathbf{\Psi}} \right) \right\|_{HS}^2 \leq \sum_{i=m+1}^p \hat{\lambda}_i^2,$$

where $\|\mathbf{A}\|_{HS} = [\text{tr}(\mathbf{A}\mathbf{A}^\top)]^{1/2}$ is the Hilbert-Schmidt norm. Here $\hat{\mathbf{L}}$ and $\hat{\mathbf{\Psi}}$ are the principal component estimators, $\hat{\lambda}_i$ is the i -th largest eigenvalue of $\mathbf{S}$. (See page 17 of Chapter 7 slides)

Hint: Since $\mathbf{S} - \hat{\mathbf{L}}\hat{\mathbf{L}}^\top - \hat{\mathbf{\Psi}}$ has zeros on the diagonal, (sum of squared entries of $\mathbf{S} - \hat{\mathbf{L}}\hat{\mathbf{L}}^\top - \hat{\mathbf{\Psi}} \leq$ (sum of squared entries of $\mathbf{S} - \hat{\mathbf{L}}\hat{\mathbf{L}}^\top$)